

Viri Blockchain Architecture

Native Account Abstraction, Dual VM & Zero-Knowledge Privacy

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The Modular Blockchain Dilemma

Why existing modular architectures introduce critical systemic risk

- **Monolithic Bottlenecks:** Single validator set handles execution, consensus, state, and DA → throughput bounded by single-node hardware limits.
- **Fragmented Modular Dependencies:** Outsourcing DA to Celestia, execution to Arbitrum, settlement to Ethereum introduces multi-hop trust layers.
- **Key Vulnerability Vectors:**
 - **Cross-Layer Bridge Risk:** Over \$2.8B exploited across third-party liquidity bridges.
 - **UX & Gas Friction:** Managing EOAs, fragmented gas tokens, ERC-4337 relayers.
 - **MEV Extraction:** Sequencers front-running & sandwiching unencrypted mempools.

The Viri Unified 3-Layer Solution

Strict layer separation without external layer dependencies

Layer 1 (Core Infrastructure): Formally verified HotStuff-2 BFT consensus, hex-radix MPT, BadgerDB persistence, libp2p authenticated networking.

Layer 2 (Execution & Privacy): Dual VM (EVM + WASM), native account abstraction (zero EOAs), Pedersen-Groth16 ZK pool, TEE MEV resistance.

Layer 3 (Application Ecosystem): On-chain Governance DAO, IBC-style inter-appchain channels, threshold validator bridge, intent solvers.

Layer 1: HotStuff-2 BFT Consensus

3-phase leader-driven BFT state machine with optimistic responsiveness

- **Quorum Condition:** Under $N \geq 3F + 1$, quorum threshold $|Q|$ satisfies:

$$|Q| \geq \left\lfloor \frac{2N}{3} \right\rfloor + 1 = 2F + 1$$

- **3-Phase Pipeline:** Linear view change complexity using Timeout Certificates (TC):

Prepare $\xrightarrow{QC_{\text{prep}}}$ PreCommit $\xrightarrow{QC_{\text{precommit}}}$ Commit $\xrightarrow{QC_{\text{commit}}}$ Decide

TLA+ Formal Safety Verification

Exhaustive TLC model checking across 34.1M+ states

Model Configuration	Total States	Distinct States	Depth	Safety Violations
$N = 4, F = 1$ (Honest)	55	32	8	0
$N = 4, F = 1$ (Byzantine)	920	412	8	0
$N = 4, F = 1$ (Full Partitions)	34,102,891	3,142,905	24	0
$N = 7, F = 2$ (Faulty Leaders)	1,429,012	284,110	16	0

Verified Invariants: Agreement, NoDoubleCommit, LockedViewInvariant, and QuorumIntersection hold across all Byzantine attack vectors.

Layer 2: Native Account Abstraction

Zero EOAs: Every account is a native smart contract wallet from genesis

- **Native UserOperation Pipeline:**

$$\text{UserOp} = \left(a_{\text{sender}}, n, \mathbf{c}_{\text{init}}, \mathbf{d}_{\text{exec}}, g_{\text{call}}, g_{\text{verify}}, a_{\text{paymaster}}, \mathbf{s} \right)$$

1. Validation Phase

Invokes $a_{\text{sender}}.\text{validateUserOp}()$. Validates session keys, social recovery signatures, and fee coverage.

2. Execution Phase

Invokes $a_{\text{sender}}.\text{executeUserOp}()$. Executes contract call state transitions upon validation success.

Dual VM & Dynamic Multi-Token Gas

Drop-in Solidity EVM + Wazero WASM runtime with dynamic EIP-1559 settlement

EIP-1559 Dynamic Base Fee

$$B_{n+1} = B_n \cdot \left(1 + 0.125 \cdot \frac{G_n - G_{\text{target}}}{G_{\text{target}}} \right)$$

Multi-Token Gas Settlement

$$\text{Fee}_K = \frac{g_{\text{used}} \cdot (B_n + p_{\text{priority}})}{\mathcal{O}(K \rightarrow \text{VIRI})}$$

- Pay transaction fees in USDC, USDT, or arbitrary ERC-20 via on-chain oracle conversion.

Pedersen-Groth16 Shielded Privacy Pool

On-chain ZK precompile over BN254 elliptic curve using gnark

Pedersen Commitment

$$C = v \cdot G + r \cdot H$$

Poseidon Nullifier

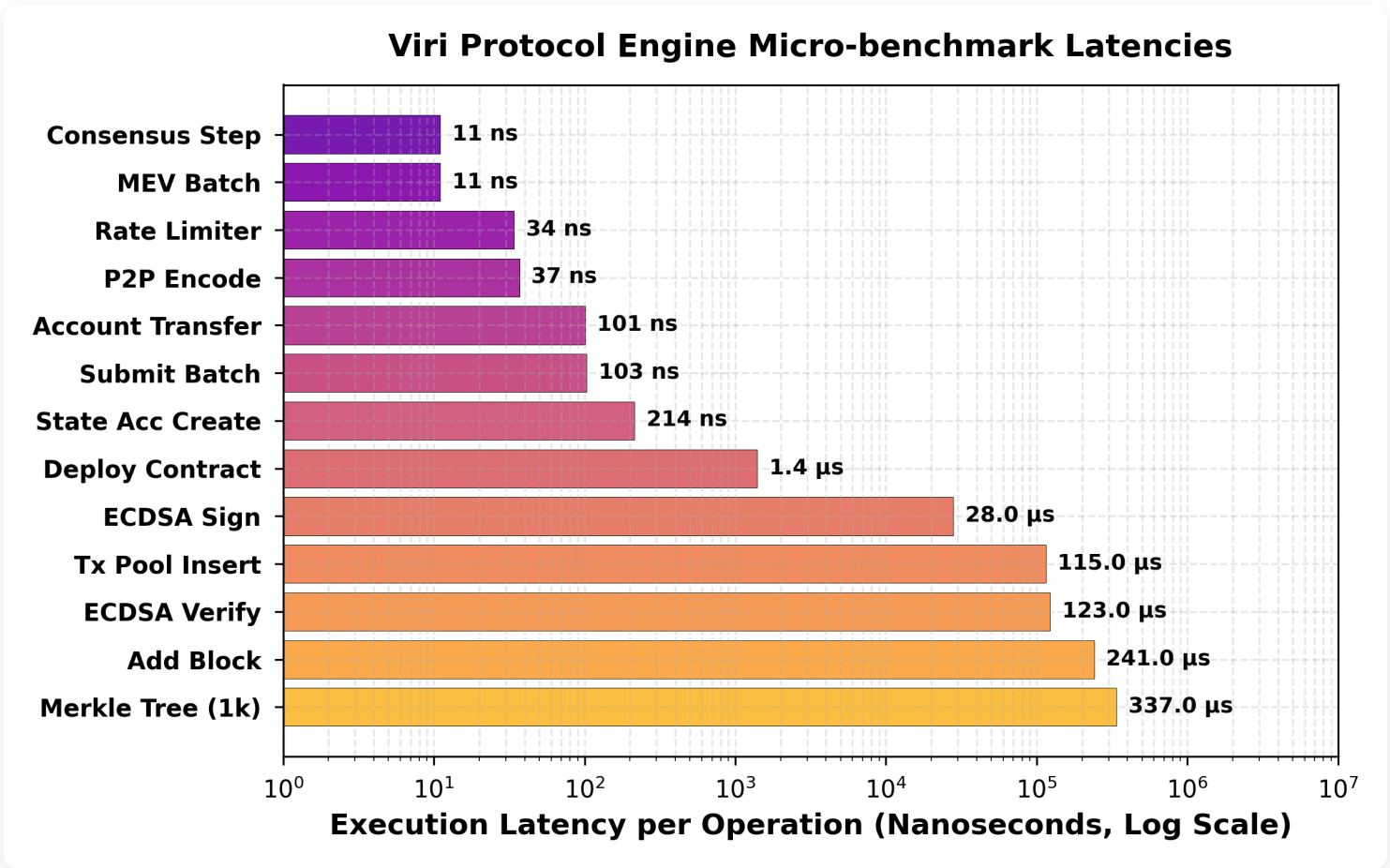
$$N = \text{Poseidon}(SK_{\text{owner}}, \text{LeafIndex})$$

Groth16 Pairing Relation Verification

$$e(\pi_A, \pi_B) = e(\alpha, \beta) \cdot e\left(\gamma_0 + \sum_{i=1}^l x_i \gamma_i, \gamma\right) \cdot e(\pi_C, \delta)$$

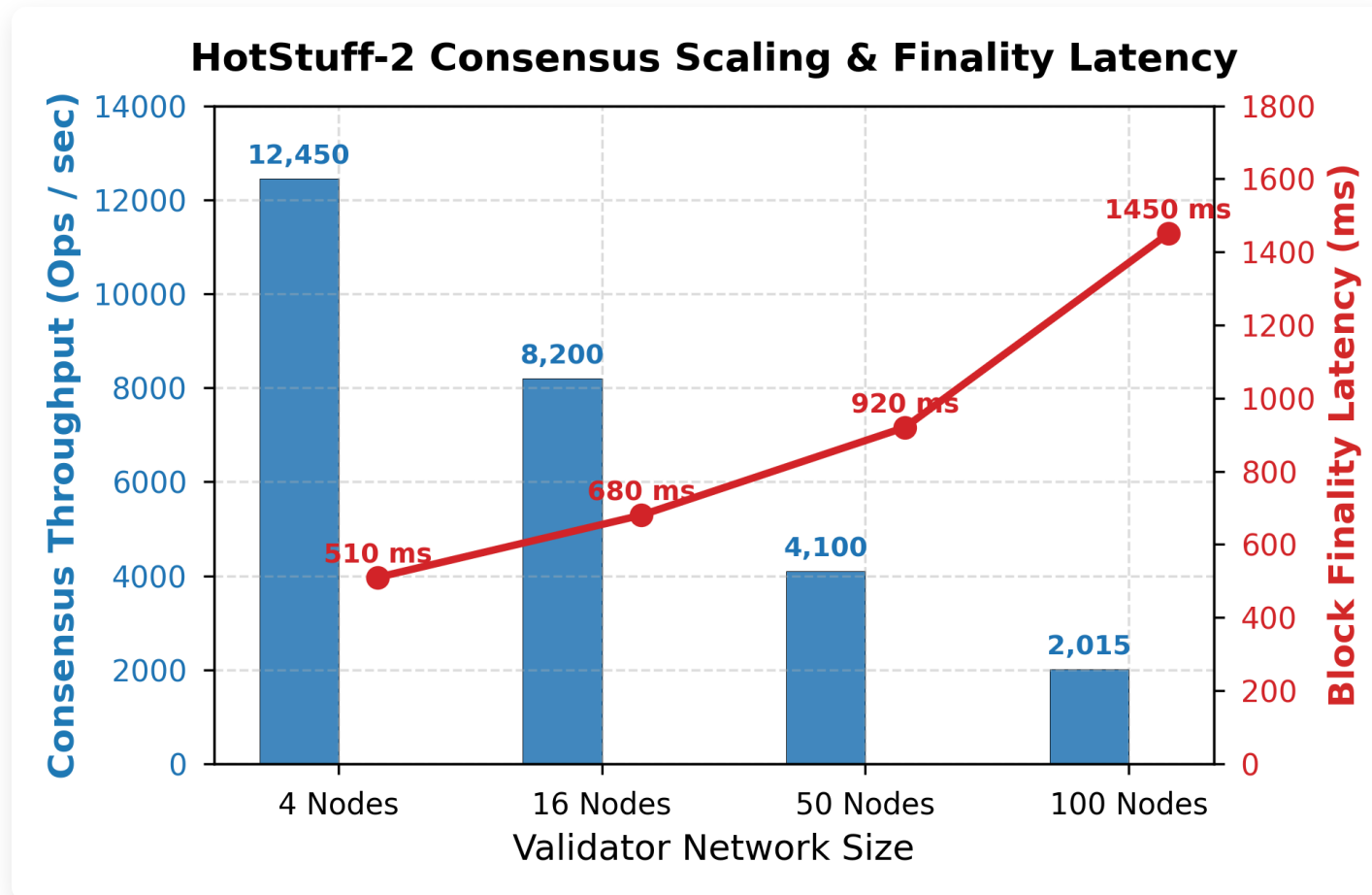
Empirical Micro-benchmark Performance

Log-scale latency breakdown across core protocol engines



HotStuff-2 BFT Consensus Scaling

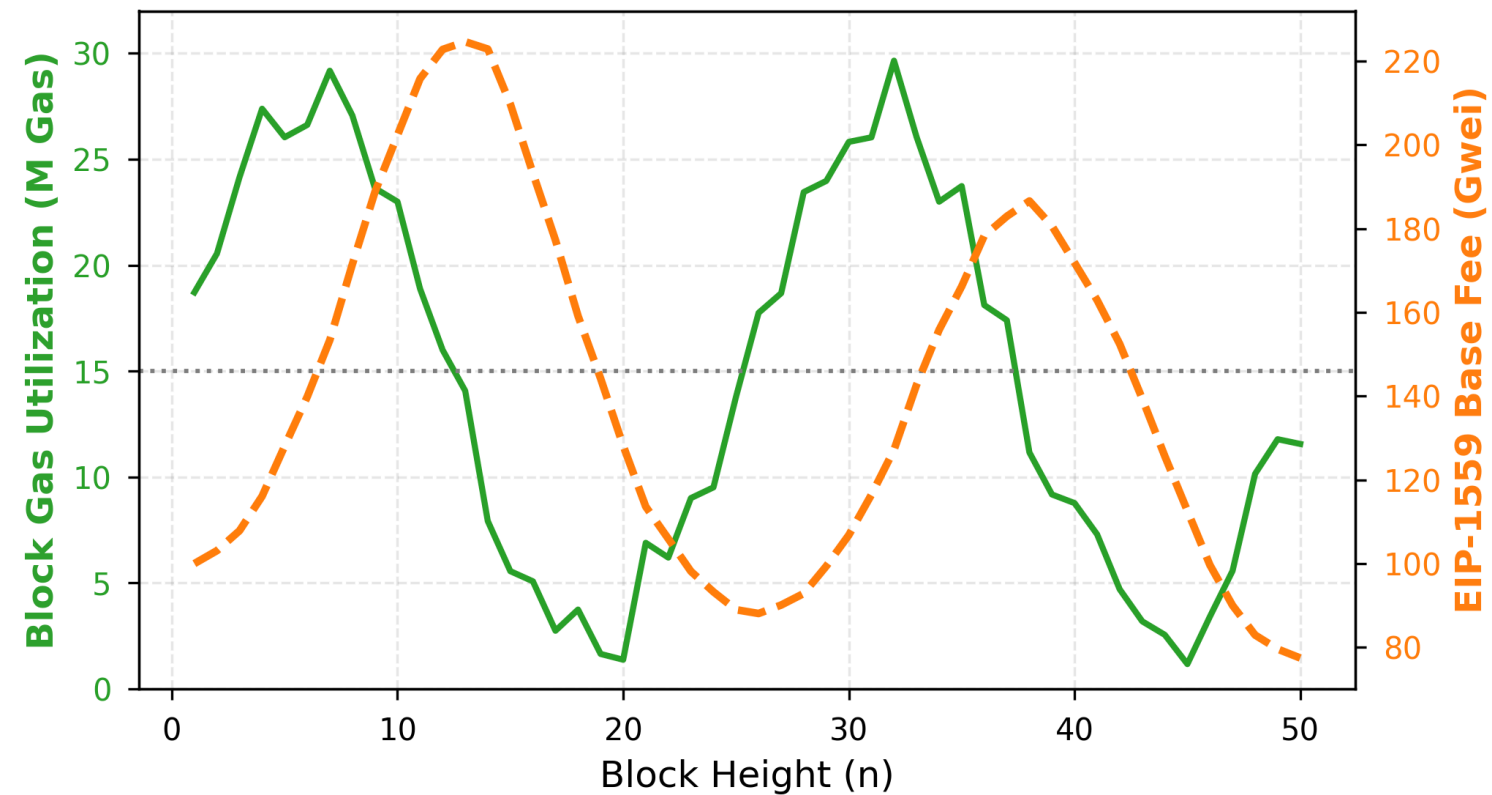
2,015 ops/sec throughput under 100-validator supermajority



EIP-1559 Dynamic Base Fee Response

Gas fee stability curve under fluctuating block utilization

Dynamic EIP-1559 Base Fee Response under Gas Demand Elasticity



Conclusion & Key Takeaways

Production-grade unified 3-layer modular blockchain

- **Unified Stack:** Eliminates external layer dependencies and bridge exploit vectors.
- **Formally Verified:** 34.1M+ TLA+ states verified with zero safety violations.
- **Developer & User Centric:** Dual EVM+WASM runtimes, zero EOAs, multi-token gas fees.
- **High Scalability & Resilience:** 2,015 ops/sec sustained under 100 validators, Jepsen fault resilient.

Project Website: <https://viri.me> • **Testnet RPC:** <https://rpc.viri.me>